INCH-POUND

MIL-PRF-39003/9E
16 December 2003
SUPERSEDING
MIL-PRF-39003/9D
17 May 1996

PERFORMANCE SPECIFICATION SHEET

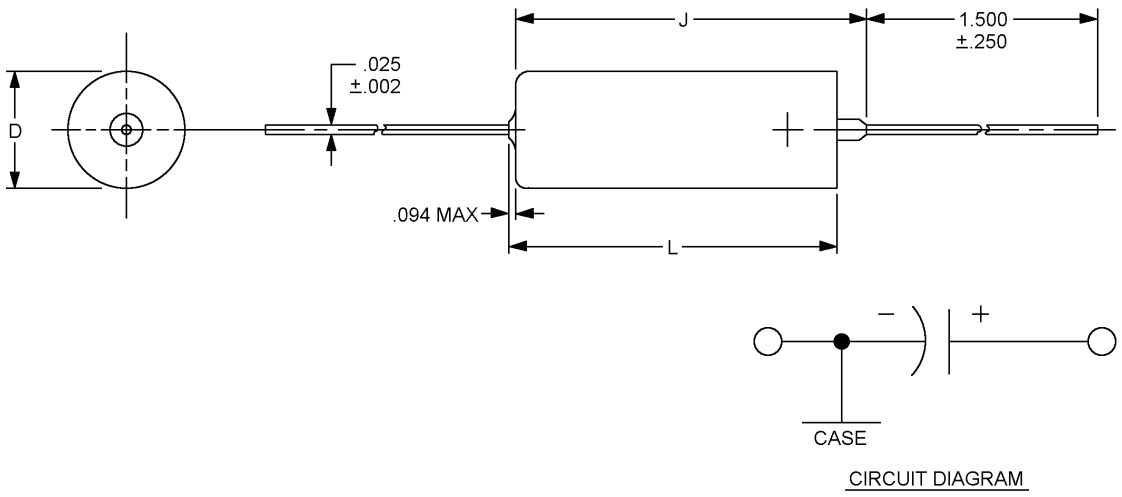
CAPACITOR, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM (POLARIZED, SINTERED SLUG), HIGH FREQUENCY, ESTABLISHED RELIABILITY, STYLE CSR21

FAILURE RATE LEVELS “M”, “P”, AND “R” ARE INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1984.

FAILURE RATE LEVEL “S” IS INACTIVE FOR NEW DESIGN AFTER 15 AUGUST 1984.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39003.



CIRCUIT DIAGRAM

Inches	mm
.002	0.05
.025	0.64
.094	2.39
.250	6.35
1.500	38.10

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Case size	Dimensions		
	L ± .031 (0.79)	D + .016 (0.41) - .015 (0.38)	J max
C	.686 (17.42)	.289 (7.34)	.822 (20.88)
D	.786 (19.96)	.351 (8.92)	.922 (23.42)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. The case insulation shall extend .015 (0.38 mm) minimum beyond each end. However, when a shrink-fitted insulation is used, it shall lap over the ends of the capacitor body.
4. Lead length may be a minimum of 1-inch long for use in tape and reel automatic insertion equipment, when specified.

FIGURE 1. Dimensions and configuration – Continued.

REQUIREMENTS:

Requirements: Requirements shall be in accordance with MIL-PRF-39003, and as specified herein.

Design and construction: Capacitors shall be of the design, construction, and physical dimensions specified on figure 1.

Case type: Tubular, insulated.

Material: Metal (alloy unspecified).

Type of seal: Hermetic.

Leads: Type N of MIL-STD-1276, tin-lead plating, electrodeposited or hot-solder dipped.

DC related voltage: See table I. Above +85°C voltage derating is required (see MIL-PRF-39003).

DC leakage (DCL): See table I.

Capacitance (Cap.) (nom): See table I.

Cap. tolerance: 5, 10, and 20 percent, see table I.

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR at 100 kHz): See table I.

Operating temperature: -55°C to +125°C.

Failure rate level: See table I.

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TABLE I. Electrical characteristics and PIN's.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at +25°C	DC leakage at +85°C	DC leakage at +125°C	Dissipation factor 1 kHz +25°C	ESR 100 kHz +25°C (max)	Ripple current 40 kHz +25°C (max)	Derated ripple current 1 kHz, +25°C (max)	Case size	PIN M39003/09- 1/ Failure rate level for (%1000 hrs.) M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
Volts	µF	Percent	µA	µA	µA	Percent	Ohms	A	A								
6	150	5	4.5	90	113	10	0.065	3.3	2	C	0001-	0101-	0201-	0301-	2001-	3001-	14001-
6	150	10	4.5	90	113	10	0.065	3.3	2	C	0002-	0102-	0202-	0302-	2002-	3002-	4002-
6	150	20	4.5	90	113	10	0.065	3.3	2	C	0003-	0103-	0203-	0303-	2003-	3003-	4003-
6	180	5	5.5	110	138	10	0.060	3.4	2.4	C	0004-	0104-	0204-	0304-	2004-	3004-	4004-
6	180	10	5.5	110	138	10	0.060	3.4	2.4	C	0005-	0105-	0205-	0305-	2005-	3005-	4005-
6	270	5	6.5	130	163	10	0.050	4.1	3.4	D	0006-	0106-	0206-	0306-	2006-	3006-	4006-
6	270	10	6.5	130	163	10	0.050	4.1	3.4	D	0007-	0107-	0207-	0307-	2007-	3007-	4007-
6	330	5	7.5	150	188	12	0.045	4.3	3.8	D	0008-	0108-	0208-	0308-	2008-	3008-	4008-
6	330	10	7.5	150	188	12	0.045	4.3	3.8	D	0009-	0109-	0209-	0309-	2009-	3009-	4009-
6	330	20	7.5	150	188	12	0.045	4.3	3.8	D	0010-	0110-	0210-	0310-	2010-	3010-	4010-
10	82	5	4	80	100	8	0.085	2.9	1.8	C	0011-	0111-	0211-	0311-	2011-	3011-	4011-
10	82	10	4	80	100	8	0.085	2.9	1.8	C	0012-	0112-	0212-	0312-	2012-	3012-	4012-
10	100	5	5	100	125	8	0.075	3	2.2	C	0013-	0113-	0213-	0313-	2013-	3013-	4013-
10	100	10	5	100	125	8	0.075	3	2.2	C	0014-	0114-	0214-	0314-	2014-	3014-	4014-
10	100	20	5	100	125	8	0.075	3	2.2	C	0015-	0115-	0215-	0315-	2015-	3015-	4015-
10	120	5	6	120	150	8	0.070	3.2	2.5	C	0016-	0116-	0216-	0316-	2016-	3016-	4016-
10	120	10	6	120	150	8	0.070	3.2	2.5	C	0017-	0117-	0217-	0317-	2017-	3017-	4017-
10	180	5	9	180	226	8	0.060	3.7	3.4	D	0018-	0118-	0218-	0318-	2018-	3018-	4018-
10	180	10	9	180	226	8	0.060	3.7	3.4	D	0019-	0119-	0219-	0319-	2019-	3019-	4019-
10	220	5	10	200	250	10	0.055	3.9	3.4	D	0020-	0120-	0220-	0320-	2020-	3020-	4020-
10	220	10	10	200	250	10	0.055	3.9	3.4	D	0021-	0121-	0221-	0321-	2021-	3021-	4021-
10	220	20	10	200	250	10	0.055	3.9	3.4	D	0022-	0122-	0222-	0322-	2022-	3022-	4022-
15	56	5	4	80	100	6	0.100	2.6	1.8	C	0023-	0123-	0223-	0323-	2023-	3023-	4023-
15	56	10	4	80	100	6	0.100	2.6	1.8	C	0024-	0124-	0224-	0324-	2024-	3024-	4024-
15	68	5	5	100	125	6	0.095	2.7	2.2	C	0025-	0125-	0225-	0325-	2025-	3025-	4025-
15	68	10	5	100	125	6	0.095	2.7	2.2	C	0026-	0126-	0226-	0326-	2026-	3026-	4026-
15	68	20	5	100	125	6	0.095	2.7	2.2	C	0027-	0127-	0227-	0327-	2027-	3027-	4027-
15	120	5	9	180	226	8	0.070	3.5	2.8	D	0028-	0128-	0228-	0328-	2028-	3028-	4028-
15	120	10	9	180	226	8	0.070	3.5	2.8	D	0029-	0129-	0229-	0329-	2029-	3029-	4029-
15	150	5	10	200	250	8	0.065	3.6	3.1	D	0030-	0130-	0230-	0330-	2030-	3030-	4030-
15	150	10	10	200	250	8	0.065	3.6	3.1	D	0031-	0131-	0231-	0331-	2031-	3031-	4031-
15	150	20	10	200	250	8	0.065	3.6	3.1	D	0032-	0132-	0232-	0332-	2032-	3032-	4032-

See footnote at end of table.

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TABLE I. Electrical characteristics and PIN's. – Continued.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at			Dissipation factor 1 kHz +25°C	ESR 100 kHz +25°C (max)	Ripple current 40 kHz +25°C (max)	Derated ripple current 1kHz, +25°C (max)	Case size	PIN M39003/09- 1/ Failure rate level for (%1000 hrs.)						
			+25°C	+85°C	+125° C						M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
Volts	µF	Percent	µA	µA	µA	Percent	Ohms	A	A								
20	27	5	2.5	50	63	5	0.145	2.2	1.2	C	0033-	0133-	0233-	0333-	2033-	3033-	4033-
20	27	10	2.5	50	63	5	0.145	2.2	1.2	C	0034-	0134-	0234-	0334-	2034-	3034-	4034-
20	33	5	3.5	70	88	5	0.130	2.3	1.4	C	0035-	0135-	0235-	0335-	2035-	3035-	4035-
20	33	10	3.5	70	88	5	0.130	2.3	1.4	C	0036-	0136-	0236-	0336-	2036-	3036-	4036-
20	33	20	3.5	70	88	5	0.130	2.3	1.4	C	0037-	0137-	0237-	0337-	2037-	3037-	4037-
20	39	5	4	80	100	5	0.120	2.4	1.7	C	0038-	0138-	0238-	0338-	2038-	3038-	4038-
20	39	10	4	80	100	5	0.120	2.4	1.7	C	0039-	0139-	0239-	0339-	2039-	3039-	4039-
20	47	5	4.5	90	113	6	0.110	2.5	1.8	C	0040-	0140-	0240-	0340-	2040-	3040-	4040-
20	47	10	4.5	90	113	6	0.110	2.5	1.8	C	0041-	0141-	0241-	0341-	2041-	3041-	4041-
20	47	20	4.5	90	113	6	0.110	2.5	1.8	C	0042-	0142-	0242-	0342-	2042-	3042-	4042-
20	56	5	5.5	110	138	6	0.100	2.9	2.2	D	0043-	0143-	0243-	0343-	2043-	3043-	4043-
20	56	10	5.5	110	138	6	0.100	2.9	2.2	D	0044-	0144-	0244-	0344-	2044-	3044-	4044-
20	68	5	7	140	175	6	0.095	3	2.4	D	0045-	0145-	0245-	0345-	2045-	3045-	4045-
20	68	10	7	140	175	6	0.095	3	2.4	D	0046-	0146-	0246-	0346-	2046-	3046-	4046-
20	68	20	7	140	175	6	0.095	3	2.4	D	0047-	0147-	0247-	0347-	2047-	3047-	4047-
20	82	5	8	160	200	6	0.085	3.1	2.5	D	0048-	0148-	0248-	0348-	2048-	3048-	4048-
20	82	10	8	160	200	6	0.085	3.1	2.5	D	0049-	0149-	0249-	0349-	2049-	3049-	4049-
20	100	5	10	200	250	8	0.075	3.3	2.5	D	0050-	0150-	0250-	0350-	2050-	3050-	4050-
20	100	10	10	200	250	8	0.075	3.3	2.5	D	0051-	0151-	0251-	0351-	2051-	3051-	4051-
20	100	20	10	200	250	8	0.075	3.3	2.5	D	0052-	0152-	0252-	0352-	2052-	3052-	4052-
35	22	5	4	80	100	4	0.160	2.1	1.5	C	0053-	0153-	0253-	0353-	2053-	3053-	4053-
35	22	10	4	80	100	4	0.160	2.1	1.5	C	0054-	0154-	0254-	0354-	2054-	3054-	4054-
35	22	20	4	80	100	4	0.160	2.1	1.5	C	0055-	0155-	0255-	0355-	2055-	3055-	4055-
35	27	5	4.5	90	113	4	0.145	2.4	1.9	D	0056-	0156-	0256-	0356-	2056-	3056-	4056-
35	27	10	4.5	90	113	4	0.145	2.4	1.9	D	0057-	0157-	0257-	0357-	2057-	3057-	4057-
35	33	5	5.5	110	138	5	0.130	2.5	1.9	D	0058-	0158-	0258-	0358-	2058-	3058-	4058-
35	33	10	5.5	110	138	5	0.130	2.5	1.9	D	0059-	0159-	0259-	0359-	2059-	3059-	4059-
35	33	20	5.5	110	138	5	0.130	2.5	1.9	D	0060-	0160-	0260-	0360-	2060-	3060-	4060-
35	39	5	7	140	175	5	0.120	2.6	2	D	0061-	0161-	0261-	0361-	2061-	3061-	4061-
35	39	10	7	140	175	5	0.120	2.6	2	D	0062-	0162-	0262-	0362-	2062-	3062-	4062-
35	47	5	8	160	200	5	0.110	2.7	2.2	D	0063-	0163-	0263-	0363-	2063-	3063-	4063-
35	47	10	8	160	200	5	0.110	2.7	2.2	D	0064-	0164-	0264-	0364-	2064-	3064-	4064-
35	47	20	8	160	200	5	0.110	2.7	2.2	D	0065-	0165-	0265-	0365-	2065-	3065-	4065-

See footnote at end of table.

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TABLE I. Electrical characteristics and PIN's. – Continued

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at			Dissipation factor 1 kHz +25°C	ESR 100 kHz +25°C (max)	Ripple current 40 kHz +25°C (max)	Derated ripple current 1kHz, +25°C (max)	Case size	PIN M39003/09- 1/ Failure rate level for (%1000 hrs.)						
			+25°C	+85°C	+125°C						M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
Volts	µF	Percent	µA	µA	µA	Percent	Ohms	A	A								
50	5.6	5	2.2	45	56	3	0.300	1.5	0.6	C	0066-	0166-	0266-	0366-	2066-	3066-	4066-
50	5.6	10	2.2	45	56	3	0.300	1.5	0.6	C	0067-	0167-	0267-	0367-	2067-	3067-	4067-
50	6.8	5	2.2	45	56	3	0.275	1.6	0.7	C	0068-	0168-	0268-	0368-	2068-	3068-	4068-
50	6.8	10	2.2	45	56	3	0.275	1.6	0.7	C	0069-	0169-	0269-	0369-	2069-	3069-	4069-
50	6.8	20	2.2	45	56	3	0.275	1.6	0.7	C	0070-	0170-	0270-	0370-	2070-	3070-	4070-
50	8.2	5	2.5	50	63	3	0.250	1.6	0.9	C	0071-	0171-	0271-	0371-	2071-	3071-	4071-
50	8.2	10	2.5	50	63	3	0.250	1.6	0.9	C	0072-	0172-	0272-	0372-	2072-	3072-	4072-
50	10	5	2.5	50	63	3	0.230	1.7	1.1	C	0073-	0173-	0273-	0373-	2073-	3073-	4073-
50	10	10	2.5	50	63	3	0.230	1.7	1.1	C	0074-	0174-	0274-	0374-	2074-	3074-	4074-
50	10	20	2.5	50	63	3	0.230	1.7	1.1	C	0075-	0175-	0275-	0375-	2075-	3075-	4075-
50	12	5	3	60	75	3	0.210	1.8	1.3	C	0076-	0176-	0276-	0376-	2076-	3076-	4076-
50	12	10	3	60	75	3	0.210	1.8	1.3	C	0077-	0177-	0277-	0377-	2077-	3077-	4077-
50	15	5	4	80	100	3	0.190	1.9	1.4	C	0078-	0178-	0278-	0378-	2078-	3078-	4078-
50	15	10	4	80	100	3	0.190	1.9	1.4	C	0079-	0179-	0279-	0379-	2079-	3079-	4079-
50	15	20	4	80	100	3	0.190	1.9	1.4	C	0080-	0180-	0280-	0380-	2080-	3080-	4080-
50	18	5	4.5	90	113	4	0.175	2.0	1.4	C	0081-	0181-	0281-	0381-	2081-	3081-	4081-
50	18	10	4.5	90	113	4	0.175	2.0	1.4	C	0082-	0182-	0282-	0382-	2082-	3082-	4082-
50	22	5	5.5	110	138	4	0.160	2.3	1.7	D	0083-	0183-	0283-	0383-	2083-	3083-	4083-
50	22	10	5.5	110	138	4	0.160	2.3	1.7	D	0084-	0184-	0284-	0384-	2084-	3084-	4084-
50	22	20	5.5	110	138	4	0.160	2.3	1.7	D	0085-	0185-	0285-	0385-	2085-	3085-	4085-

1/ Dash number shall include the letter "H" to indicate an internal construction solder with a minimum melting point of 221°C, or the "H" shall be deleted.

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REQUIREMENTS – Continued:

Stability at low and high temperatures: In accordance with MIL-PRF-39003.

Step 1 (+25°C):

DCL: See table I.

Cap.: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

ΔCap.: Shall not change more than ± 10 percent of step 1 measured value.

DF: 2 times step 1 measured value or table I, whichever is greater.

Step 3 (+25°C):

DCL: See table I.

ΔCap.: Shall not change more than ± 2 percent of step 1 measured value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap.: Shall not change more than ± 8 percent of step 1 measured value.

DF: 1.5 times step 1 measured value or table I, whichever is greater.

Step 5 (+125°C):

DCL: See table I.

ΔCap.: Shall not change more than ± 12 percent of step 1 measured value.

DF: 1.5 times step 1 measured value or table I, whichever is greater.

Step 6 (+25°C):

DCL: See table I.

ΔCap.: Shall not change more than ± 2 percent of step 1 measured value.

DF: See table I.

Surge voltage: In accordance with MIL-PRF-39003.

DCL: See table I.

ΔCap.: Shall not change more than ± 2 percent of initial measured value.

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DF: See table I.

Surge current: This test shall be performed on a 100 percent basis following thermal shock and voltage aging on initial qualification and group A tests. Capacitors shall be subjected to 3 consecutive cycles of the following conditions:

Temperature: +25°C ± 5°C.

Applied voltage: Rated dc voltage ± 2 percent from a power source having an energy storage bank of 50,000 µF (minimum) across the output terminals.

Charge cycle: 4 ± 1 seconds.

Discharge cycle: 4 ± 1 seconds to a voltage below 1 percent of rated voltage.

The total dc resistance (excluding the capacitor) including the wiring, fixturing and output impedance of the regulated power supply to each test position during the charging cycle shall be 1 ± 0.2 ohm.

On the completion of the test, the capacitors shall meet the following requirements:

DCL: See table I.

ΔCap.: Shall be as specified in table I.

DF: See table I.

Life: Method 108 of ML-STD-202.

240, 1,000, and 2,000 hours:

At +25°C (as applicable):

DCL: 200 percent of initial limit.

ΔCap.: Shall not change more than ± 5 percent of initial measured value.

DF: See table I.

ESR: (after 240 and 2,000 hours): See table I.

At +85°C:

DCL: See table I.

At +125°C:

DCL: See table I.

4,000, 6,000, 8,000, and 10,000 hours (exponential only):

At +25°C (as applicable):

DCL: 400 percent of initial limit.

ΔCap.: Shall not change more than ±10 percent of initial measured value.

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DF: See table I.

ESR: (after 10,000 hours): See table I.

At +85°C:

DCL: 200 percent of initial limit.

At +125°C:

DCL: 200 percent of initial limit.

Marking: In accordance with MIL-PRF-39003.

Part or Identifying Number (PIN): M39003/09-(dash number from table I).

APPLICATION NOTES:

1. Rated ripple current is the rms value of the maximum allowable alternating current of a specified frequency, at which the capacitor may be operated continuously at a specified temperature. Derate ripple current for ambient temperature in accordance with the curve shown on figure 2.

2. For derating the frequency, use the derated ripple currents at 1 kHz given in table I. Below 1 kHz these same currents are applicable provided the peak ac voltage does not result in voltage reversal or exceeds the rated dc voltage. Between 1 kHz and 40 kHz the ripple current may be interpolated linearly with frequency. The ripple current is applicable at frequencies above 40 kHz.

3. Although style CSR21 capacitors are rated to operate with the specified levels of rms ripple current, they are basically polar devices. Care must be exercised to assure that sufficient dc bias is applied to prevent ac voltage reversal in excess of specified reverse voltage ratings (see MIL-HDBK-198).

4. When two or more style CSR21 capacitors are used in parallel, ripple current may not divide equally as a result of unequal ESR's of the capacitors. It is imperative that each capacitor be operated within the specified limit of rms ripple current.

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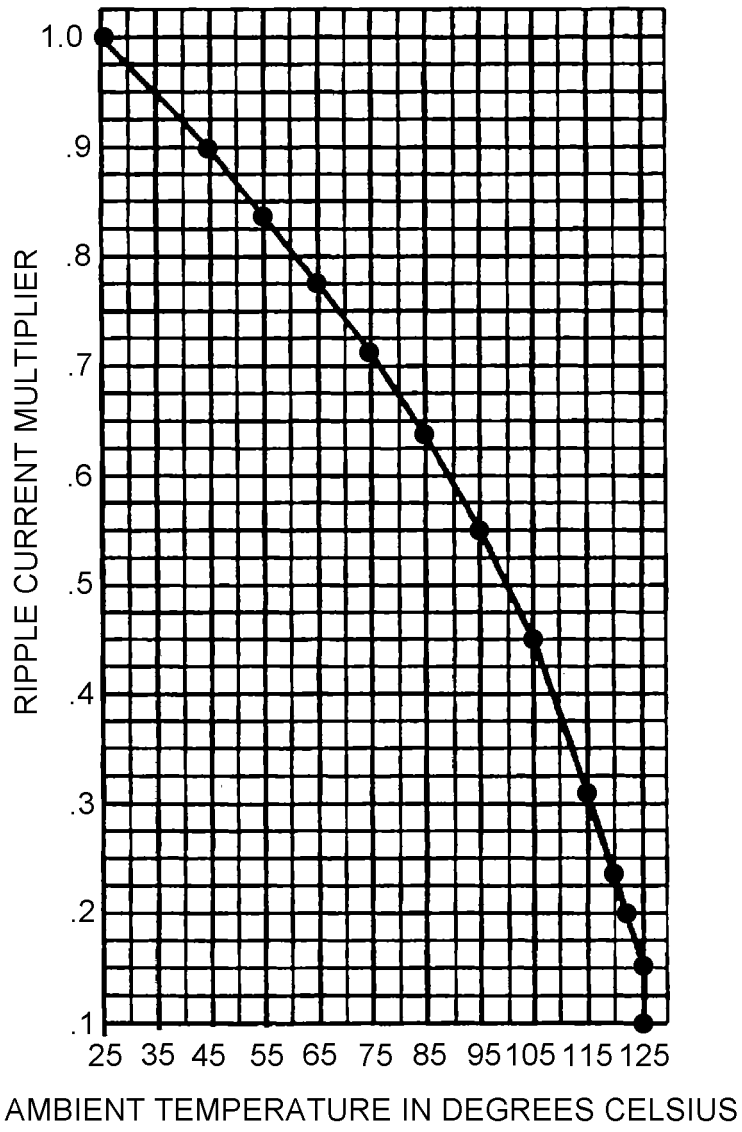


FIGURE 2. Ripple current derating with respect to temperature.

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Substitutability data: Substitutability for failure rate levels and capacitance tolerances shall be in accordance with MIL-PRF-39003.

* NOTES:

Reference documents. In addition to MIL-PRF-39003, this specification sheet references the following documents.

DEPARTMENT OF DEFENSE

MIL-STD-202 - Test Method Standard, Electronic and Electrical Component Parts.

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Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA - CC

Review activities:
Navy – AS, MC, OS, SH
Air Force – 19, 99

Preparing activity:
DLA – CC

(Project 5910-2203-03)

* NOTE: the activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil .

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